Fluency Calculator

Participatory activities for purposeful, transparent, and people-centric dataset documentation

Fluency Calculator

Prerequisites checklist

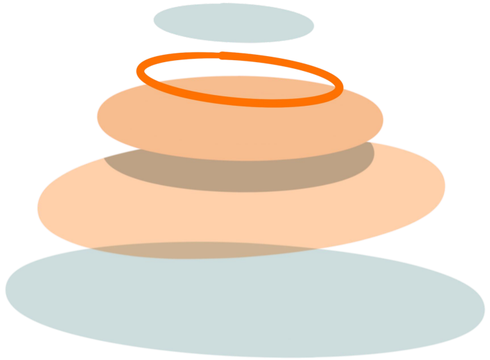
BEFORE YOU BEGIN

–

→ Update agent proficiency in the “Agents Brainstorm” tab of the Activity Tracker

→ Update the required proficiency for dimensions in the “Input: Dimensions Worksheet” tab of the Activity Tracker

→ Have you data card or template ready to inspect



Fluency Calculator

Data Cards should be readable by agents with the lowest fluency in data.

- Flag Dimensions that require a fluency greater than the lowest fluency rating
- Add or remove questions, decide on trade-offs, and discuss how to write answers that agents can easily understand

Follow the steps to use the Fluency Calculator

Step 1: Rank the Dimensions by importance to agents in the Input: Dimensions Ranking (for each Agent) tab in the Activity Tracker

Step 2: As a group, review the Output: Calculators tab in the Activity Tracker

Step 3: Note action items, improvement strategies, and refinements. Track in Dimensions Worksheets.

Step 4: Refine your template and update lenses/scopes in the Activity Tracker

These can be empirically validated once the Data Cards have been filled out.

Making discussions actionable

The colors in the Calculators indicate how easy it might be for agents to read your Data Card.

Use the following prompts to structure discussion, and create a plan to refine your template.

- Which questions will agents find most helpful when making assessments about the dataset for their use cases?
- What information might be missing?
- How granular is the information solicited from the questions? Is this sufficient for agents?
- What additional resources can we point to?

👎 **Accountability**

↑ DEFINITELY DIFFICULT

👍 Risk and Recommendations

↑ LIKELY EASY

👍 Use and Utility

↑ LIKELY EASY

🤔 Impact and Consequences

↑ COULD BE DIFFICULT

👎 **Quality**

↑ DEFINITELY DIFFICULT

#datacardsplaybook

Activity — Fluency Calculator

	Agent 1	Agent 2	Agent 3	Agent 4	Agent 5
1	Copy–paste dimension from sticker sheet here.				
2					
3					
4					
5					

ACCOUNTABILITY	RISK & RECOMMENDATIONS	USE & UTILITY	IMPACT & CONSEQUENCES	QUALITY

Use the sticker-sheet below to copy and paste dimensions in the table.

Decide how important each dimension is to your agents.

- Color the box Red if the required fluency for the dimension is more than the available fluency of the agent. These boxes require special attention.
- Color the box Yellow if the required fluency for the dimension is within the range of the available fluency of the agent. These will require some attention.
- Color the box Green if the required fluency for the dimension is well below the available fluency of the agent.

Do this automatically in the Activity Tracker. Ensure that your agents' fluency ratings are captured in the Agents Brainstorm tab. Update the Input: Dimensions Ranking (for each agent) tab to see the results in the Output: Fluency Calculators Tab.

Pro-tips for Fluency Calculator

🦁 **Help agents make assessments**
The goal is to have agents plausibly evaluate all Dimensions

🐇 **Keep on moving**
If you get stuck, move to the next Dimension and come back later

💬 **Reference personas frequently**
The Persona Worksheets cover the benefits, tasks, and uses for each agent

❓ **Sleuth out evidence**
Look for examples of questions in your template as potential evidence

🗞️ **Ruthlessly eliminate**
Use an elimination strategy if you are unable to prioritize

Next Steps

OUTCOMES

Directional insights and areas for improvement for your Data Card or documentation.

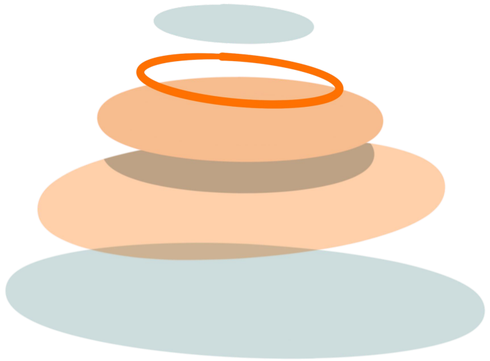
A clear understanding of which agents might find your dataset's documentation challenging to read, and why that might be ok.

FOLLOW UP

→ Fine-tune your Data Card or Template

→ Parking lot: Take note of action items or improvements

→ Next: OFTEn Calculator



The Data Cards Playbook ↗ by Google Research ↗ is licensed under a Creative Commons Attribution-ShareAlike 4.0 International License. You are free to share and adapt this work under the appropriate license terms ↗.